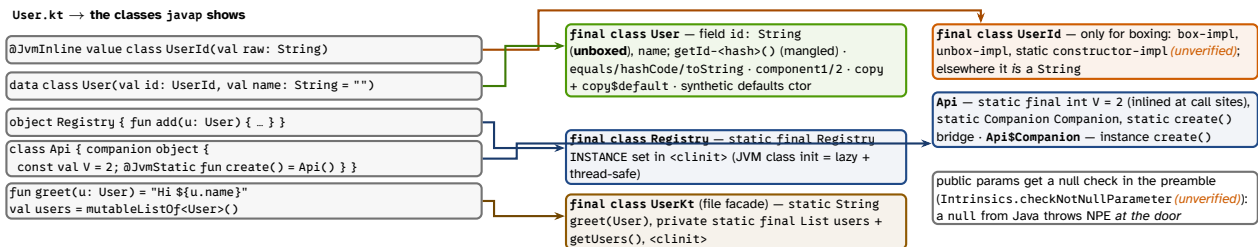


In one line: The JVM knows nothing about Kotlin: no top-level functions, no properties, no companions, no default arguments, no value classes, no suspend. **kotlinc** lowers each feature to plain classes, static methods, synthetic bridges and runtime helper calls (kotlin.jvm.internal.Intrinsics). Knowing the lowering explains Java-interop annotations, boxing costs, \$default frames in stack traces, and why a read-only List can still change.

User.kt → the classes javap shows



See it yourself: IDE Tools → Kotlin → Show Kotlin Bytecode → Decompile, or javap -c -p.
@file:JvmName("Users") renames the facade; internal members get name\$module names.

Declarations → JVM

- **Properties** = private field + getX()/setX() (isX stays isX). @JvmField exposes the field; const val (primitive/String, top-level/object/companion) is a compile-time constant **copied into callers** — change it, recompile them.
- **Companion** = nested class Outer\$Companion + static field; its members are *instance* methods. @JvmStatic adds a real static in Outer, so Java writes Api.create().
- **Default args**: one body + synthetic f\$default(args..., int mask, Object); each bit of mask says “use the default”. Java sees only the full signature — add @JvmOverloads for telescoping overloads.
- **Lambdas**: invokedynamic + LambdaMetafactory by default since 2.0 (smaller APKs; not serializable; -XLambdas=class reverts). Captured vars become Ref.IntRef/ObjectRef.
- **lateinit var**: non-null field; the getter null-checks and throws UninitializedPropertyAccessException; ::x.isInitialized; no primitives. A nullable var instead makes every read ?.!!.
- No checked exceptions in Kotlin: without @Throws(IOException::class) Java cannot catch (IOException e) around the call — “never thrown in try”.

Example — Java-facing API

```

class Client @JvmOverloads constructor(
    val url: String, val retries: Int = 3, val gzip: Boolean = true) {
    companion object {
        const val VERSION = 2 // Java: Client.VERSION (inlined)
        @JvmField val TAG = "client" // Java: Client.TAG (static field)
        @JvmStatic fun local() = Client("https://example.com")
        fun plain() = 1 // Java: Client.Companion.plain()
    }
    @Throws(IOException::class) fun fetch(): ByteArray = TODO()
}
val a: Int? = 1000; val b: Int? = 1000
a == b // true -> equals()
a === b // false -> two Integer boxes (cache is -128..127)
  
```

Control flow & collections

- when (s) on **sealed/is**: an instanceof chain; on JVM 21+ a single invokedynamic SwitchBoostraps.typeSwitch (experimental 2.2.20, **default since 2.4.20**). On an **enum**: synthetic \$WhenMappings ordinal table + tableswitch (unverified).
- **Read-only ≠ immutable**: List is a compile-time view of java.util.List; listOf(a, b) is Arrays.asList-backed (unverified), mutableListOf an ArrayList — a cast or Java code can mutate it. setOf iterates in insertion order.
- **Iterable** ops are eager — every step builds a new list. **Sequence** is lazy, *per element* (map 1 → filter 1 → ...), short-circuits (first); pays off for long chains/big data, not 10 items or sorted().

Boxing

- Int → int; Int?, List<Int>, generic T → Integer. IntArray = int[], Array<Int> = Integer[]. == on boxes compares objects.

Compiler & targets

- **K2** (stable, default since 2.0, May 2024): new frontend **FIR** feeds a shared **IR** backend → JVM bytecode, JS, Wasm, or LLVM (Native).
- **Kotlin/Native**: LLVM machine code, a **tracing GC** (new memory manager, default since 1.7.20, no freeze); ARC only at the ObjC/Swift edge.

Traps

- ! “List is immutable” — only a read-only *interface*.
- ! foo\$default in a trace = the default-args bridge.
- ! Changed a library const val: callers keep the *old* inlined value.

Remember: FileKt · INSTANCE · Outer\$Companion · \$default + mask · indy lambdas · List = java.util.List.